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Explainable Job-Posting Recommendations Using Knowledge Graphs and Named Entity Recognition

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***Abstract*— The growth of online job-posting repositories provided job-seekers with access to a large number of potential jobs. User assessment of recommended jobs becomes especially a tedious and time-consuming task with the overwhelming number of job recommendations. To enhance the job-seeker's ability to evaluate the suitability of a recommended job, we propose an explainable job recommendation system, which matches the user to the most relevant jobs based on their profile. Then, the system explains to the user why each job-posting has been recommended to them. The proposed system uses a knowledge graph (KG) structure to model job-postings and user profiles in one homogeneous structure. Graph relations between the job-seekers and job-postings are mined through natural language processing (NLP) of the textual content from job-postings and user-profiles. Based on the graph structure itself and a customized named entity classifier, a human-readable explanation is generated for each recommendation and provided to the job-seeker. The explanation includes information about the matching factors that led the system to recommend a certain job-posting to the user. The proposed system is implemented and tested on a sample data-set of user profiles and job-postings from open online repositories. We use BELU and Rouge-L scores to show that the proposed systems generated relevant explanations for recommended jobs.**

I. INTRODUCTION

Intelligent recommender systems have shown great potential in recent years for supporting human decision-making across a wide range of domains and applications. One main factor that played an important role in supporting the abilities of intelligent recommenders is the availability of large volumes of data about users and the recommended items. Such data included mainly information about user preferences and their ratings of the items. This enabled the recommender system of learning how to match, with high accuracy, a certain user to items they will most likely accept. However, this high accuracy is also bound to the number of available ratings in any data set. If the user-item rating matrices are largely sparse, the accuracy of the recommender is considerably impacted [1]. This is known as the sparsity problem. It is especially noticeable in the case where new users are introduced to system, while no previous ratings or preferences exist for them, leading to the cold-start problem [1].

In the context of job recommendations, the rating matrix sparsity and cold-start problems become of a high influence when the user does not or cannot provide ratings of the job-postings. In the majority of job-posting repositories, users cannot rate the job-posting. Even in cases where rating a job-posting is possible, the postings remain in the system only until the job opening is occupied, where it will be then deleted. This rapid and dynamic change in the available job-postings, and the continuous introduction of new users looking for openings, requires different strategies to create job recommendations.

Several solutions have been suggested in the literature to handle the highlighted issues in recommending job-postings to users [2], [3], [4]. Among those solutions, knowledge-graph-based recommendations gained an increasing interest in the recent years due to the knowledge graph's ability to create an interlinked network of users and job-posting features. Those networks form a strong foundation for recommending, to the user, jobs that are "connected" to them. The connections between jobs and users are constructed as relations within the knowledge graph. Graph relations are generated based on mining several types of similarities between users and job-posting. Traditional methods need user ratings or information about the selection of products or jobs. Since applying this mechanism is very limited within job portals, we are proposing a new method of finding similarities between users and job-postings based on knowledge graphs. Furthermore, knowledge graphs are introduced to express the relationships between users and jobs in an explainable way, as well as to generate a base structure to extract graph-based similarity ratings. The explanations are a direct result of the embedded contextual knowledge about users and job-postings in the knowledge graph. Therefore, such explanations can be represented in a human-readable form, to support the user in understanding the reasoning behind each job recommendation.

In this paper, we build on the concepts of multidimensional knowledge representation in [5], and graph-based recommendation approaches [6], to propose an explainable job-posting recommender system, which offers the user a comprehensible justification for the recommended jobs. In the proposed system, we implement multiple NLP techniques on the textual content of job-postings and user profiles to calculate the similarities between a certain user and the potential jobs that are suitable for them. Jobs and users are represented as nodes in the knowledge graph. Graph relations are then created based on the calculated similarities. Recommendations are generated from the direct and transitive relations in the knowledge graph. To generate a high-level explanation for the user, we create textual templates with information slots, which are filled with the needed information from graph relations and the NLP methods. We propose a concept for the creation of template slots based on

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the classification of entities in the textual data. A customized named entity recognition (NER) has been developed and tailored towards the job-market taxonomy. This allowed the textual explanations to be more understandable to the user and more accurate in their representation of the reasoning of the recommender system.

In the following sections of this paper, we present an overview of the related literature in section 2. In section 3, we discuss the proposed explainable graph-based recommender system. The system's evaluation and results are discussed in section 4, and the work is concluded with future perspectives in section 5.

II. Related Work

Recommender systems have been widely used in industry and well investigated in academia. This is due to the wide spectrum of their applications in real life. Generally, three types of recommendation algorithms dominated the literature on recommender systems [6], [7]: content-based recommender systems, collaborative filtering (CF) recommender systems, and hybrid recommender systems. The hybrid recommendation approach was developed to overcome the shortcomings of both CF-based and content-based recommender [8]. Those problems emerged from the core logic of both recommender types, which depends, almost solely, on the available data about the users and the recommended items. The introduction of knowledge graphs and semantic web showed that recommender systems can also utilize the context of the data in generating more accurate recommendations, through mapping the mutual relations between users and items in the KG [9]. Moreover, the semantics of the KG can enable generating a comprehensible explanation of the recommended items [10], based on embedded relations between users and items in the KG, which are constructed based on predefined explainable criteria.

In the job-market domain, recommender systems have been developed to support employers and employees in finding relevant candidates and job-postings respectively. Job recommender systems use the available information about the job-seekers, e.g. from their CVs or profiles, as well as information about the job from the job-posting. Recommendation algorithms ranged along the spectrum of traditional, hybrid, and knowledge-based ones. In the work of Zhang et al. [11], authors have used a CF approach to recommend the top three related jobs to the user. Their approach depends on vector space models to investigate the similarities between users and job-postings. Graph-based job recommendation systems began as an approach to provide hybrid recommendations, i.e. CF- and content-based ones, such as in the early work of Huang et al. [12]. In the recent years, this approach became more advanced and capable of harvesting the potentials of knowledge graphs. For example, Shalaby et al. [13] constructed a job recommendation system on a knowledge graph structure that uses multiple types of relations between the graph nodes. Since the nodes represent users and jobs, the multi-type relations allowed them to capture and mode multiple behavioral and contextual similarities between users and jobs. Gugnani et al. [14] addressed in their graph-based recommendations the importance of considering candidate's skills in generating job recommendations for certain domains with non-standardized job roles. They constructed a skill-graph and use it as the foundation of the job recommendation system.

Previous approaches, among several others, focused on the accuracy of the recommendation algorithm. However, in an application are like job recommendations, the explainability of the recommended jobs have also been under the spotlight. Zhang et al. [15] point out in their survey that most of the ongoing explainable recommender systems use the user's preferences and item descriptions through various machine learning models. They treated the recommendation model as a black-box and develop a parallel model that explains the results of the recommendation. Those black-box explanations still require extra information on user preferences and item descriptions. Guo et al. [6] highlight in their survey a class of explainable recommender systems, which intersects with several other classes of recommendation algorithms, especially the graph-based, ontology-based, or semantic-web approaches. Examples of such directions can be seen in the works of Song et al. [16], where the authors use a sequential decision process to generate explainable KG-based recommendations. Ribeiro et al. [17] discussed that explaining an output through textual or visual artifacts provides a qualitative understanding of the relationship between the user's query and the model's output. Elsafty et al. [18], investigated the use of dense vector representations to enhance a large-scale job recommendation system. They ranked German-based job-postings in terms of their similarities, which provided the needed document embeddings. Wang et al. [19] developed an explainable approach of document recommendations based on knowledge graphs. They utilize higher-order relations between the connected users and items in the KG to enhance the recommendations. Their approach, namely Knowledge Graph Attention Network (KGAT), uses attention mechanisms to distinguish the importance of node's neighbors. We build on the explainability approach in [19] and extend it with a more detailed, template-based, explanation model. Our approach defines the information slots and their requirements in the explanation template. It retrieves the needed information from graph nodes and relations and extends those with a dedicated NLP-based skill-classification module, which is used to generate comprehensible explanations in the job-posting recommendation domain, alongside its role in the construction of the KG itself.

III. Explainable, Knowledge-Graph Based, Job Recommendations

The proposed explainable job-recommendation system in this work is designed through a multi-stage process, shown in Fig.1, which includes the following main steps:

1) Data collection and processing.
2) Document embedding generation, where the NLP model is employed to calculate the semantic similarities between documents.

3) Knowledge graph construction, where job-postings and user profiles are modeled with their connections in the form of nodes and relations in the graph structure. This structure is then used as the knowledge base for generating the recommendation.
4) Explanation construction, where a customized named entity classifier is implemented to provide the needed values for a template-based textual explanation. Moreover, visual explanations are provided from the graph structure itself.

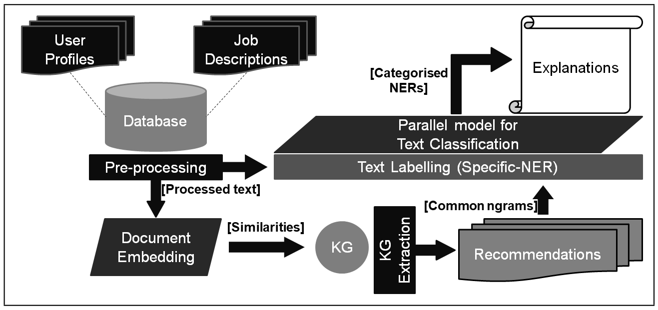

Fig. 1. The diagram shows the architecture of the proposed methodology. Through the knowledge graph, the direct and indirect recommendations are provided, and with the help of a parallel explainability model the template-based explanation is given.

A. Document Embedding for Semantic Similarities

Document embedding is the technique to predict the context of words and sentences in a given document, through mapping them into a vector space that can represent their semantic information. This method was developed by Mikolov and Le in 2014 [20]. The authors found that the Paragraph Vector - Distributed Memory (PV-DM) model outperformed the sentence prediction models. Therefore, we have chosen and implemented this model to provide the semantics of job-posting and user-profile documents.

Originally, this process follows the "Word2Vec" method, developed by Mikolov in 2013 [21]. The difference is that the paragraph's ID or document's ID is inserted in the document embedding model along with the input query, to locate the exact document from the corpus. Later, these document vectors are mapped into space to identify the similarities between them using cosine similarity (1).

$$\cos(\theta)_{(A_{i,j},B_{i,j})} = \frac{\sum_{i,j}^{n} S_{i,j} \cdot A_i \cdot B_j}{\sqrt{\sum_{i,j}^{n} S_{i,j} \cdot A_i \cdot A_j} \cdot \sqrt{\sum_{i,j}^{n} S_{i,j} \cdot B_i \cdot B_j}} \tag{1}$$

here, $\mathbf{S}$ is the feature similarity matrix, where $S_{i,j}$ = similarity(i,j). The formula states that "As the angle between the documents increases, the less similar the documents would be". As shown in Fig. 2, the angle θ between UP2 and JD document is less than the one between JD and UP1, which results in user profile 2 being more relevant to the job description. Similarly, the angle between UP1 and UP2: Γ is about 90 degrees which means that the UP1 and UP2 documents are completely different.

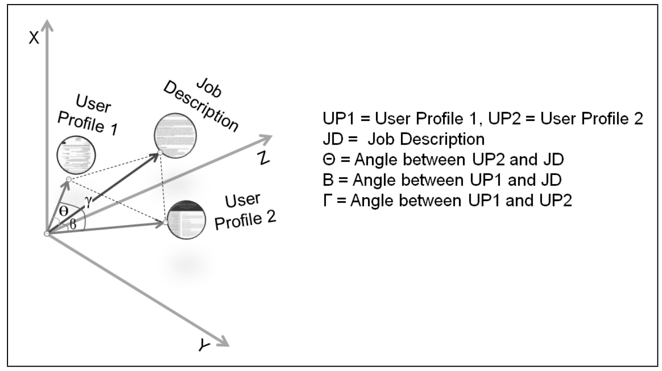

Fig. 2. Similarity measure on document vectors

B. Knowledge Graph Construction

Once document similarities are calculated, we map every document into the graph as a node. Each node in the graph represents a user profile or a job posting. Relations between the nodes are created from the corresponding similarity between the documents they represent. Mathematically, the knowledge graph is defined as Graph G = Set of (N, R), where, N is a set of nodes $x, y = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$; and $R \subseteq \{\{x, y\} | x, y \in N$ and $x \neq y\}$, is a set of edges between two the nodes.

Following this definition, we create the knowledge graph to include three types of document relations: userProfile-userProfile, userProfile-jobPosting, and jobPosting-jobPosting. This structure enables the use of a CF approach to generate job recommendations. CF-based recommendations in this context are accomplished based on the user groups, which include user profiles with the highest similarities, and job-posting groups that include job-postings with the highest similarities.

To implement the CF-based recommender, we use Dijkstra's shortest path algorithm [22] in a way that it finds the intermediate node between the source node, which is a user-profile, and the target node, which is a job-posting in our use-case. With an intermediate node, two relations link the source and target nodes transitively, following the transitive property of equality: if $a = b$ and $b = c$ then $a = c$. In our case, the equality is replaced with the similarity between the nodes: if $a \sim b$ and $b \sim c$ then $a \sim c$.

Two recommendation types are then generated from the knowledge graph; 1) Direct recommendations that provide the user with the most similar job postings from the nodes that are directly connected to the user profile node. 2) CF-based recommendations, which are job postings that are recommended to a certain user based on user and job-posting groups in the knowledge graph, as shown in Fig. 3. The user receives job recommendations through one of the two transitive paths: 1) userProfile (A) $\rightarrow$ userProfile (B) $\rightarrow$ jobPosting, which is generated from the group of users (A). 2) userProfile $\rightarrow$ jobPosting (A) $\rightarrow$ jobPosting (B), which is generated from the group of job-posting (A).

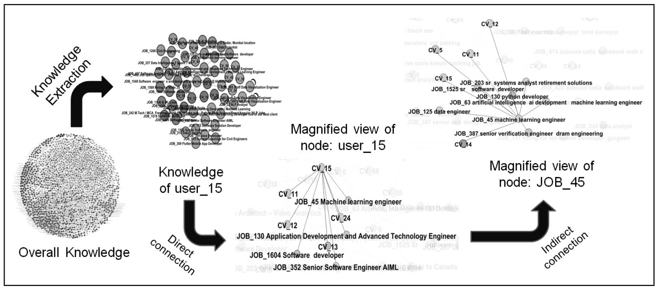

Fig. 3. The diagram shows the overview of the knowledge graph recommendations: Here, the knowledge of user profile 15 is extracted, in which user profile 15 is directly connected to four profiles and four job descriptions. These directly connected profiles and jobs are again connected directly to other jobs and profile (i.e. job 45), which are indirectly recommended for the user profile 15.

C. Graph-based Explainability

Our proposed solution builds on the approaches shown in [20] and extends them by defining a model of textual and graphical explanations through the knowledge graph. Explanations are provided with each recommended job-posting in a human-comprehensible form. The proposed explainability method uses different sentence templates for direct recommendations and CF-based ones. Those templates, see Fig.4, are built by filling information slots with respective information from the similarity measurements, the graph structure, and a customized name entity classifier.

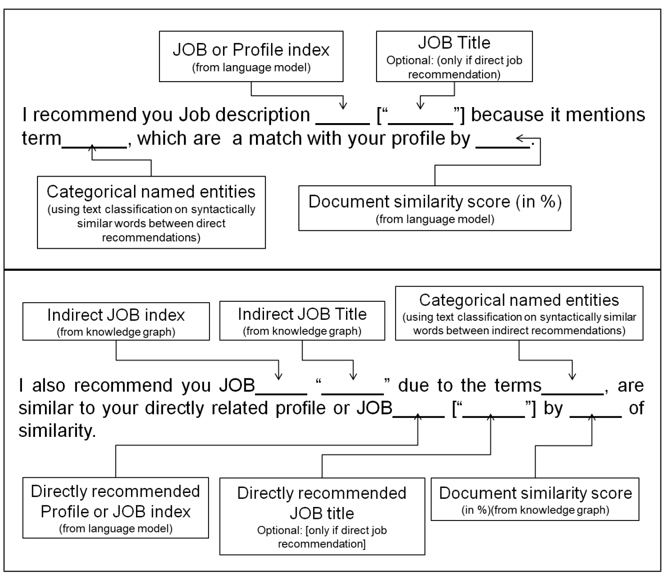

Fig. 4. Sentence template for the direct (above) and extended (below) recommended job explanation.

The first explanation template uses syntactically similar words to fill the information slots with similar terminology between the user profile and the job posting. It then provides the user with the actual similarity score to the recommended job. The second template is equipped with more information through implementing a customized name entity classifier on the language model. In this template, the user gets more specific information about the categories that related their user profile to a certain job posting.

We selected six entity categories that are relevant to the information technology (IT) and engineering sector in our evaluation, which successfully enhance the content and clarity of the explanation. To achieve our customized named entity classifier, we developed a data set of more than 60,000 unique words with 8 custom labels, including "Study", "Software", "Programming_language", "Skills", "Programming_framework", "location", "job_title" and "other_entity" categories. A Support Vector Classifier (SVC) is utilized for this task.

The explanation algorithm implements three functions to generate the recommendation' explanation:

- Extract the syntactical similar terms from the source profile and recommended job description.
- classify these similar terms using the customized NER.
- Segregate the similar terms according to their labels into the labeled terms to fill the explanation template.

With this algorithm, the generated explanation sentence will look like the following example:

> *I recommend you Job number **509 "Software Engineer"** because it is similar to you through: '**Study**': ['science', 'analysis', 'control', 'bachelor', 'automate', 'data'], '**Software**':['oracle'], '**programming_language**': ['java', 'python'], '**Skills**': ['agile', 'research', 'development', 'experience', 'software', 'application'], Other similar terms to this job are: ['architecture', 'server', 'product', 'technical', 'framework', 'design', 'technology'], which are a match with your profile by **51.682**%.*

The explanation is designed to be informative and specific. This explanation is also supported by the information from user and job-posting groups, where two more information pieces are added to the template: the similar user or job-posting, and the similarity score of the transitive relation.

IV. Results and Discussion

A. Data Set Preparation

Our proposed methodology is tested in the IT and engineering domain. We construct the data set of user profiles and job postings through collecting job descriptions from the job-hunting website "Monster"[1]. Since the EU General Data Protection Regulation (GDPR)[2] applies constraints on to use of personal information, we use a set of anonymous user profiles that has been constructed from sample CVs in the IT and engineering sectors. This data set has been constructed as an educational example and thus does not include personal information.

B. Similarities Between User Profiles and Job Descriptions

Our experiment was conducted with 50 user profiles and 1606 IT- and engineering-related job postings. After applying the language model on user profile data, maximum, median, and minimum similarity scores between documents were

[1] https://www.monsterindia.com/
[2] https://gdpr-info.eu/art-5-gdpr/

calculated to determine the variance of document similarities. An example of these scores from two job postings are illustrated in Fig.5. From the similarity distribution analysis,

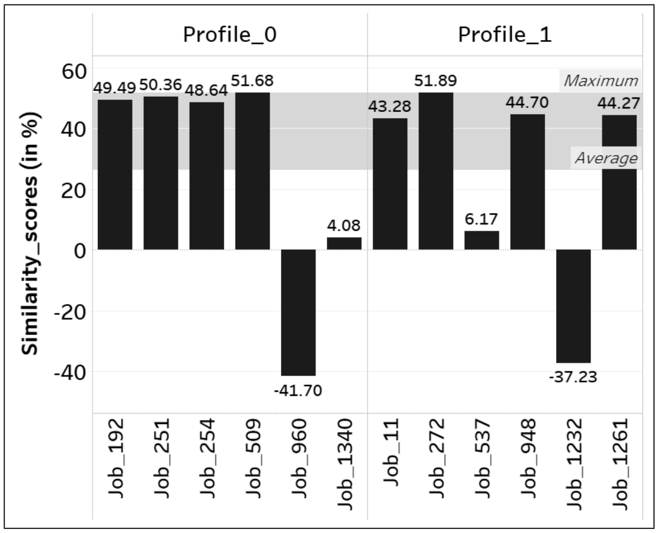

Fig. 5. Job description similarities to two user profiles, with the maximum and average similarity distribution. The Job-posting 509 is similar to user profile 0. Similarly job-posting 272 is suitable to user profile 1, whereas job postings 960 and 1232 are completely different from user profiles. The negative sign indicates the opposite direction in vector space.

we select the threshold value of >40% (Slightly above average distribution) and recommend the top 4 similar job-postings to the respective user profile.

C. Knowledge Graph Evaluation

The knowledge graph is a graphical database containing the information of users, job postings, and their potential similarities. This information is extracted through navigating graph nodes through their respective relations. We refer to navigating through one relation as a ”jump”. Our experiment shows the the higher number of jumps in the graph, the lower the relevance is to the target node.

TABLE I

RELEVANCY OF JOB RECOMMENDATIONS TO THE USER PROFILE WITH DIFFERENT NUMBER OF JUMPS ON THE KNOWLEDGE GRAPH

	Recommended job and its similarity (in %) with user profile		
	n_jump = 1	**n_jump = 2**	**n_jump = 3**
User Profile 0	Job_509 51.68	Job_429 37.15	Job_226 25.16
User Profile 1	Job_272 51.89	Job_1085 39.29	Job_1084 20.51
User Profile 2	Job_61 52.27	Job_127 38.18	Job_330 20.69

Table 1 shows how the job similarities decrease as the number of jumps increases through the knowledge graph. For example, in with one jump, user profiles 0, 1, and 2 are directly recommended the Job_509, Job_272, and Job_61 respectively. As the number of jumps increases, the job relevancy decreases. Therefore, we select a jump value of 2 to provide the collaborative filtering recommendations to the user, assuring a high relevancy score.

D. SVC Classifier Results

The SVC was utilized on custom NER, which is trained on a data set with more than 60,000 unique words. The classifier takes 70% data for the training and 30% data for the testing. We achieve 88% of test accuracy in predicting the correct label of a specific term. Table 2 shows the three main measures: recall, precision, and f1-measure, for our entity classes in our customized classifier.

TABLE II

PRECISION, RECALL, AND F1 SCORES OF OUR SVC-BASED CLASSIFIER PERFORMANCE ON CUSTOM NER

	precision	**recall**	**f1-score**	**support**
Programming_framework	0.84	0.85	0.84	241
Skills	0.79	0.88	0.83	2610
Software	0.79	0.90	0.85	902
Study	0.87	0.86	0.86	4481
job_title	0.89	0.99	0.94	480
location	0.99	0.96	0.98	485
other_word	0.93	0.88	0.90	9190
programming_language	0.92	0.98	0.95	642
accuracy			0.88	19031

E. Explainability Evaluation

We follow two methods to evaluate the explainability approach: first by evaluating the performance of the recommendation model, and second by evaluating the quality of the generated explanation template [23]. The textual explanation relies on the syntax and grammar of the sentence. Therefore, with the second method, we calculate the BELU and Rouge-L scores [24] on different explanations of respective recommendations. To calculate the scores, the user profile and its recommendations' content are combined for reference, and the generated explanation is fed as a test sentence. The results are shown below in Fig.6 for multiple profiles and job-postings. In Fig.6, the lower Recall and F1 measure scores

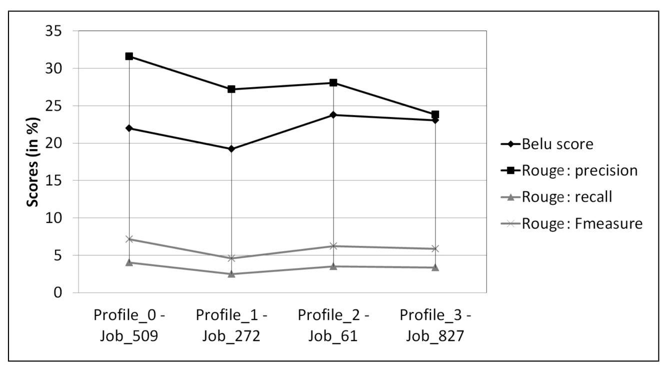

Fig. 6. BELU and Rouge-L scores on various explanations for user profiles and recommended jobs.

are caused by the fixed part of the template structure. Words of the explanation template such as ”recommend”, ”you”, ”because”, etc., are not present in the reference sentence (user profile and recommended job description). Therefore, these unrelated words affect the overall BELU and Rouge-L scores. According to [25], the BELU score and Rouge-L precision score are in a good range if it is above 20%, which shows that the generated explanations are related to the recommended job descriptions. Our BELU and Rouge-L scores ranged around an average of 26%.

V. CONCLUSION AND FUTURE WORK

In this paper, we proposed a graph-based, explainable recommender system for job-postings. The proposed system used NLP to analyze the textual content of job-postings and user profiles. We calculate the similarities between users and job-postings, as well as between users amongst each other, and job-postings amongst each other as well. User-user, job-job, and user-job similarities are embedded in a knowledge graph structure, which formed the foundation for generating the explainable recommendations. Explanations were generated using a template-based approach, where the information is inserted in the template's slots after being mined from the graph structure and a customized NER algorithm. The customized NER has been developed based on a job-market taxonomy. The proposed system has been evaluated quantitatively and qualitatively. The evaluation results showed an enhanced recommendation accuracy in comparison to normal keyword matching. The presence of explainability with the recommended jobs enhanced the use of the overall recommender system, which is more likely to increase the acceptance of the recommended jobs.

In future work, we will extend the evaluation process with a questionnaire approach, in order to include other users with different user evaluation criteria such as their votes on recommendation and explanation satisfaction, etc. Furthermore, the explanation will also be enhanced by supporting the textual templates with a visual explanation aspect, where users can directly see the relation between their recommendations and explanations in the knowledge graph.

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